

Convergencia en probabilidad implica la existencia de una subsucesión que converge casi seguro

Proposición 1. Sea $(X_n)_{n \in \mathbb{N}}$ tal que $X_n \xrightarrow[n \rightarrow \infty]{\mathbb{P}} X \implies \exists (X_{n_k})_{k \in \mathbb{N}} : X_{n_k} \xrightarrow[k \rightarrow \infty]{c.s.} X$.

Demostración: Definimos $\forall k \in \mathbb{N} : n_k = \min \{n \in \mathbb{N} : \mathbb{P}(|X_n - X| > 2^{-k}) \leq 2^{-k}\}$ y llamaremos $A_k := \{|X_{n_k} - X| > 2^{-k}\}$. Como $\mathbb{P}(|X_n - X| > \varepsilon) \xrightarrow{n \rightarrow \infty} 0$, $\forall k \in \mathbb{N} : \exists n_k \in \mathbb{N}$

$$\implies \sum_{k=1}^{\infty} \mathbb{P}(A_k) \leq \sum_{k=1}^{\infty} 2^{-k} = 1 < \infty.$$

Entonces, por el Lema de Borel-Cantelli I, $\mathbb{P}\left(\limsup_{k \rightarrow \infty} A_k\right) = 0$.

Por otro lado, afirmamos que $\left\{\omega \in \Omega : \limsup_{n \rightarrow \infty} |X_{n_k}(\omega) - X(\omega)| \neq 0\right\} \subset \limsup_{k \rightarrow \infty} A_k$.

En efecto, si $\omega \in \Omega$ es tal que $\limsup_{n \rightarrow \infty} |X_{n_k}(\omega) - X(\omega)| \neq 0$, entonces $\exists$ infinitos índices k tales que $\omega \in A_k$. Por tanto, $\omega \in \limsup_{k \rightarrow \infty} A_k$. ■

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